

DESIGN REPORT

Elevated Reinforced Concrete Water Tank

Project Name	EWT-001
Location	Nagpur, Maharashtra
Tank Type	Circular (Cylindrical) Flat-Bottom
Capacity	800 m³
Date	15 June 2026
Design Code	IS 3370:2009 IS 456:2000 IS 875 IS 1893 Part 2
Design Method	Working Stress Method (WSM)
Status	

1. Tank Type Optimisation & Selection

All feasible tank types were designed for the given capacity and site conditions. The optimum type is selected based on minimum cost per unit capacity and structural adequacy.

Rank	Tank Type	Volume (m³)	Cost/m³ (Rs.)	Concrete (m³)	Steel (kg)	Design OK	Recommendation
#1	Circular	327.6	Rs.6,859	327.6	30860	OK	Excellent economical & structural
#2	Intze	494.4	Rs.11,125	494.4	50452	OK	Good adequate design

2. Input Parameters & Tank Geometry

Parameter	Value
Internal diameter, D	12.750 m
Internal radius, R	6.375 m
Water depth, H	6.500 m
Overall height	6.800 m
Provided capacity	829.9 m ³
Free board	0.300 m
Staging height	12.000 m
Number of columns	6

3. Component-Wise Design Calculations

[OK] Cylindrical Wall

Parameter	Value	Unit
Thickness	250	mm
H^2/Dt parameter	13.255	–
Max hoop tension, T_{hoop}	371.4	kN/m
Base moment, M_{base}	158.9	kN·m/m
Hoop steel, A_{st}	2,122.0	mm ² /m
Hoop bar dia	12	mm
Hoop spacing	105	mm
Hoop steel provided	2,154.2	–
Vertical steel, A_{st}	4,770.4	mm ² /m
Vertical bar dia	20	mm
Vertical spacing	65	mm
Vertical steel provided	4,833.2	–

[OK] Top Dome (Roof)

Parameter	Value	Unit
Rise	2.550	m
Dome radius, R_d	9.244	m
Half-angle	43.600	°
Design load, w_0	4.000	kN/m ²
Meridional thrust, N	21.450	kN/m
Dome thickness	100	mm
Dome steel, A_{st}	300.0	mm ² /m
Dome bar dia	10	mm
Dome bar spacing	200	mm
Top-ring tension, T	99.000	kN
Top-ring steel, A_{st}	565.7	–
Top-ring bar dia	12	mm
Top-ring spacing	150	mm

[OK] Flat Floor Slab

Parameter	Value	Unit
Thickness	800	mm
Floor pressure	70.020	kN/m ²
Centre moment, M	569.1	kN·m/m
Steel, A_{st}	4,516.3	mm ² /m
Bar dia	32	mm
Bar spacing	150	mm
Steel provided	5,361.7	–

[OK] Bottom Ring Beam

Parameter	Value	Unit
Width, b	300	mm
Depth, d	1275	mm
Total load, W	12,767.4	kN
Moment, M	1,775.7	kN·m
Steel, A_{st}	8,570.3	mm ²
Bar dia	25	mm
Bar spacing	55	mm

[OK] Staging Columns & Bracings

Parameter	Value	Unit
Column diameter	1500	mm
Effective length, L_{eff}	10.200	m
Slenderness ratio	6.800	–
Column load, P	4,043.0	kN
No. of longitudinal bars	105	–
Longitudinal bar dia	16	mm

4. Seismic & Wind Analysis

4.1 Seismic Design — IS 1893 Part 2:2016 Two-Mass Model

The elevated tank is modelled as a two-degree-of-freedom system with impulsive (m_i) and convective (m_c) liquid masses per IS 1893 Part 2.

Parameter	Value
Seismic zone	IV
Zone factor, Z	0.240
Importance factor, I	1.500
Response factor, R	2.500
h/D ratio	0.510
Impulsive mass, m_i	0.306 m
Convective mass, m_c	0.694 m
Impulsive mass, m_i	244.4 t
Convective mass, m_c	555.5 t
Structural mass, m_s	834.9 t
Impulsive period, T_i	4.601 s
Convective period, T_c	3.823 s
S_a/g (impulsive)	0.340
S_a/g (convective)	0.356
Design coeff., $A_{h,i}$	0.025
Design coeff., $A_{h,c}$	0.026
Impulsive shear, V_i	259.2 kN
Convective shear, V_c	139.6 kN
Base shear, V_B	294.4 kN
Impulsive moment, M_i	4,258.1 kN·m
Convective moment, M_c	2,485.4 kN·m
Overturning moment, M_{ot}	4,930.4 kN·m
Lateral stiffness, K_s	2,012.5 kN/m

4.2 Wind Force — IS 875 Part 3:2015

Parameter	Value
Design wind speed, V_z	47.000 m/s
Wind pressure, p_z	1.325 kN/m ²
Force on tank, F_{tank}	80.440 kN
Force on staging, F_{stg}	23.380 kN

Wind base shear, V	103.8 kN
Wind moment, M	1,563.3 kN·m

4.3 Foundation & Global Stability

Isolated footing per column (WSM); status: **ADEQUATE**.

Parameter	Value
Service column load, P	4,043.0 kN
Overturning increment	257.8 kN
Max column load, $P_{\max}$	4,300.8 kN
Min column load, $P_{\min}$	3,785.2 kN
Footing size (square)	5.250 m
Footing thickness	600 mm
Effective depth, d	542 mm
Bearing pressure	171.6 kN/m ²
Allowable SBC	187.5 kN/m ²
Moment at column face, M	274.3 kN·m/m
Steel, A_{st}	2,991.9 mm ² /m
Bar dia	16 mm
Bar spacing	65 mm
Steel provided	3,093.2 mm ² /m
Development length, L_d	317 mm
Punching shear (allowable)	0.880 N/mm ²
One-way shear stress	0.384 N/mm ²
FoS, overturning	31.370
FoS, sliding	41.200
Net uplift?	No

5. Reinforcement Summary

Component	Steel Type	Dia (mm)	Spacing (mm)	Ast Reqd (mm²/m)	Ast Prov (mm²/m)
Wall	HYSD	12	105	2,122.0	2,154.2
Top Dome	HYSD	—	—	—	—
Floor Slab	HYSD	32	150	4,516.3	5,361.7
Bottom Ring Beam	HYSD	25	—	—	—
Staging	HYSD	—	—	—	—

6. Bar Bending Schedule (BBS)

Unit weight = $d^2 / 162.2$ kg/m (where d = bar dia in mm). Lengths include standard hooks and laps per IS 2502.

Mark	Location	Dia (mm)	Shape	Cut Len (m)	Nos	Total Len (m)	Weight (kg)
W1	Wall – Hoop (outer)	12	Straight (ring)	40.655	130	5285.19	4692.15
W2	Wall – Vertical	20	Straight	7.3	1232	8993.6	22179.04
D1	Dome – Circumferential	10	Circular ring	8.411	2	16.822	10.37
D2	Dome – Circumferential	10	Circular ring	16.422	2	32.844	20.25
D3	Dome – Circumferential	10	Circular ring	24.433	2	48.866	30.13
D4	Dome – Circumferential	10	Circular ring	32.444	2	64.888	40.01
D5	Dome – Circumferential	10	Circular ring	40.455	2	80.911	49.88
F1	Floor – Radial	32	Straight	6.675	267	1782.225	11251.53
F2-1	Floor – Circumferential	32	Ring	0.866	2	1.732	10.93
F2-2	Floor – Circumferential	32	Ring	1.797	2	3.595	22.69
F2-3	Floor – Circumferential	32	Ring	2.729	2	5.458	34.45
F2-4	Floor – Circumferential	32	Ring	3.66	2	7.321	46.22
F2-5	Floor – Circumferential	32	Ring	4.592	2	9.184	57.98
F2-6	Floor – Circumferential	32	Ring	5.523	2	11.047	69.74
F2-7	Floor – Circumferential	32	Ring	6.455	2	12.91	81.5
F2-8	Floor – Circumferential	32	Ring	7.386	2	14.773	93.26
F2-9	Floor – Circumferential	32	Ring	8.318	2	16.636	105.03
F2-10	Floor – Circumferential	32	Ring	9.249	2	18.499	116.79
F2-11	Floor – Circumferential	32	Ring	10.181	2	20.362	128.55
F2-12	Floor – Circumferential	32	Ring	11.112	2	22.225	140.31
F2-13	Floor – Circumferential	32	Ring	12.044	2	24.088	152.07
F2-14	Floor – Circumferential	32	Ring	12.976	2	25.951	163.83
F2-15	Floor – Circumferential	32	Ring	13.907	2	27.814	175.6
F2-16	Floor – Circumferential	32	Ring	14.839	2	29.677	187.36
F2-17	Floor – Circumferential	32	Ring	15.77	2	31.54	199.12
F2-18	Floor – Circumferential	32	Ring	16.702	2	33.403	210.88
F2-19	Floor – Circumferential	32	Ring	17.633	2	35.266	222.64
F2-20	Floor – Circumferential	32	Ring	18.565	2	37.129	234.4
F2-21	Floor – Circumferential	32	Ring	19.496	2	38.992	246.17
F2-22	Floor – Circumferential	32	Ring	20.428	2	40.855	257.93
F2-23	Floor – Circumferential	32	Ring	21.359	2	42.718	269.69
F2-24	Floor – Circumferential	32	Ring	22.291	2	44.581	281.45
F2-25	Floor – Circumferential	32	Ring	23.222	2	46.444	293.21
F2-26	Floor – Circumferential	32	Ring	24.154	2	48.307	304.97

Mark	Location	Dia (mm)	Shape	Cut Len (m)	Nos	Total Len (m)	Weight (kg)
F2-27	Floor – Circumferential	32	Ring	25.085	2	50.17	316.74
F2-28	Floor – Circumferential	32	Ring	26.017	2	52.034	328.5
F2-29	Floor – Circumferential	32	Ring	26.948	2	53.897	340.26
F2-30	Floor – Circumferential	32	Ring	27.88	2	55.76	352.02
F2-31	Floor – Circumferential	32	Ring	28.811	2	57.623	363.78
F2-32	Floor – Circumferential	32	Ring	29.743	2	59.486	375.54
F2-33	Floor – Circumferential	32	Ring	30.674	2	61.349	387.31
F2-34	Floor – Circumferential	32	Ring	31.606	2	63.212	399.07
F2-35	Floor – Circumferential	32	Ring	32.537	2	65.075	410.83
F2-36	Floor – Circumferential	32	Ring	33.469	2	66.938	422.59
F2-37	Floor – Circumferential	32	Ring	34.4	2	68.801	434.35
F2-38	Floor – Circumferential	32	Ring	35.332	2	70.664	446.11
F2-39	Floor – Circumferential	32	Ring	36.263	2	72.527	457.88
F2-40	Floor – Circumferential	32	Ring	37.195	2	74.39	469.64
F2-41	Floor – Circumferential	32	Ring	38.127	2	76.253	481.4
F2-42	Floor – Circumferential	32	Ring	39.058	2	78.116	493.16
F2-43	Floor – Circumferential	32	Ring	39.99	2	79.979	504.92
B1	Bottom Ring Beam – Main	25	Straight (ring)	40.655	4	162.621	626.62
B2	Bottom Ring Beam – Stirrups	10	Rectangular stirrups	0.35	201	673.35	415.14
C1L	Col 1 – Longitudinal	16	Straight	12.8	105	1344.0	2121.23
C1T	Col 1 – Ties	8	Circular ring	5.012	61	305.756	120.64
C2L	Col 2 – Longitudinal	16	Straight	12.8	105	1344.0	2121.23
C2T	Col 2 – Ties	8	Circular ring	5.012	61	305.756	120.64
C3L	Col 3 – Longitudinal	16	Straight	12.8	105	1344.0	2121.23
C3T	Col 3 – Ties	8	Circular ring	5.012	61	305.756	120.64
C4L	Col 4 – Longitudinal	16	Straight	12.8	105	1344.0	2121.23
C4T	Col 4 – Ties	8	Circular ring	5.012	61	305.756	120.64
C5L	Col 5 – Longitudinal	16	Straight	12.8	105	1344.0	2121.23
C5T	Col 5 – Ties	8	Circular ring	5.012	61	305.756	120.64
C6L	Col 6 – Longitudinal	16	Straight	12.8	105	1344.0	2121.23
C6T	Col 6 – Ties	8	Circular ring	5.012	61	305.756	120.64
	TOTAL						63857.2 kg

7. Abstract of Cost Estimate

Based on DSR 2023-24 rates (indicative; for detailed estimate use BOQ).

#	Item of Work	Qty	Unit	Rate (Rs.)	Amount (Rs.)
01	Concrete for tank walls (incl. formwork & curing)	68.1	m ³	Rs.8,000	Rs.544,720
02	Concrete for domes / roof	14.8	m ³	Rs.8,500	Rs.125,885
03	Concrete for floor slab	102.1	m ³	Rs.7,500	Rs.766,050
04	Concrete for ring beams / girder	15.3	m ³	Rs.8,000	Rs.122,560
05	Concrete for staging columns	127.2	m ³	Rs.7,500	Rs.954,225
06	Steel reinforcement (supply & place)	30860.0	kg	Rs.75	Rs.2,314,500
07	Waterproofing (2-coat epoxy, water face)	260.4	m ²	Rs.350	Rs.91,140
08	MS ladder, access platform, manhole	1.0	LS	Rs.109,745	Rs.109,745
09	Foundation (isolated/raft; assumed)	1.0	LS	Rs.658,469	Rs.658,469
10	Water testing & commissioning	800.0	m ³	Rs.50	Rs.40,000
	Sub-Total				Rs.5,727,294
	Contingency @ 5%				Rs.286,365
	GST @ 18%				Rs.1,082,459
	GRAND TOTAL				Rs.7,096,117

8. Code References & Notes

- [1] IS 3370 (Part I–IV):2009 – Concrete structures for storage of liquids
- [2] IS 456:2000 – Plain & Reinforced Concrete – Code of Practice
- [3] IS 875 (Part 1–3):1987/2015 – Dead, Live & Wind Loads
- [4] IS 1893 (Part 2):2016 – Seismic design of liquid retaining tanks
- [5] SP 16:1980 – Design Aids for Reinforced Concrete to IS 456
- [6] All calculations are based on Working Stress Method (WSM) as mandated by IS 3370.
- [7] Minimum concrete grade: M25 for liquid-retaining members (IS 3370 Part I, cl. 4).
- [8] Cover to water face: 45 mm; other face: 25 mm (IS 3370 Part I, cl. 4.1).
- [9] Permissible crack width: 0.1 mm (severe) / 0.2 mm (moderate) – IS 3370 Part I, cl. 8.1.
- [10] This report is for preliminary/detailed design; independent verification is recommended.